

Merged Intrusion Dataset 7 Percent - Data Analysis Report

Dataset: D:\Tai lieu UIT\Tai lieu ky

6\IDPS\Project\merged_intrusion_dataset_7_percent_canonical.csv

Output folder: D:\Tai lieu UIT\Tai lieu ky

6\IDPS\Project\merged_intrusion_dataset_7_percent_canonical_analysis

Generated at: 2026-05-16 22:04:06

Key findings

- Total rows analyzed: 6,264,391; number of classes: 45.
- Majority class is TFTP with 1,405,780 rows (22.4408%).
- Minority class is Infiltration with 1 rows (0.0000%).
- Majority/minority imbalance ratio: 1405780.00:1.
- Effective number of classes: 13.33 out of 45 actual classes.
- Labels below 1 percent: DDOS attack-HOIC, DDoS attacks-LOIC-HTTP, DoS attacks-Hulk, UDP-lag, Bot, DoS Hulk, FTP-BruteForce, SSH-Bruteforce, Portmap, Infiltration, PortScan, DoS attacks-SlowHTTPTest, DDoS, DoS attacks-GoldenEye, DoS attacks-Slowloris, DoS GoldenEye, FTP-Patator, DoS Slowhttpstest, SSH-Patator, DoS slowloris, UDPLag, DDOS attack-LOIC-UDP, Web Attack Brute Force, Web Attack XSS, Brute Force -Web, WebDDoS, Brute Force -XSS, SQL Injection, Infiltration, Web Attack Sql Injection.
- Heatmap computed from 120,000 sampled rows and 20 selected numeric features.

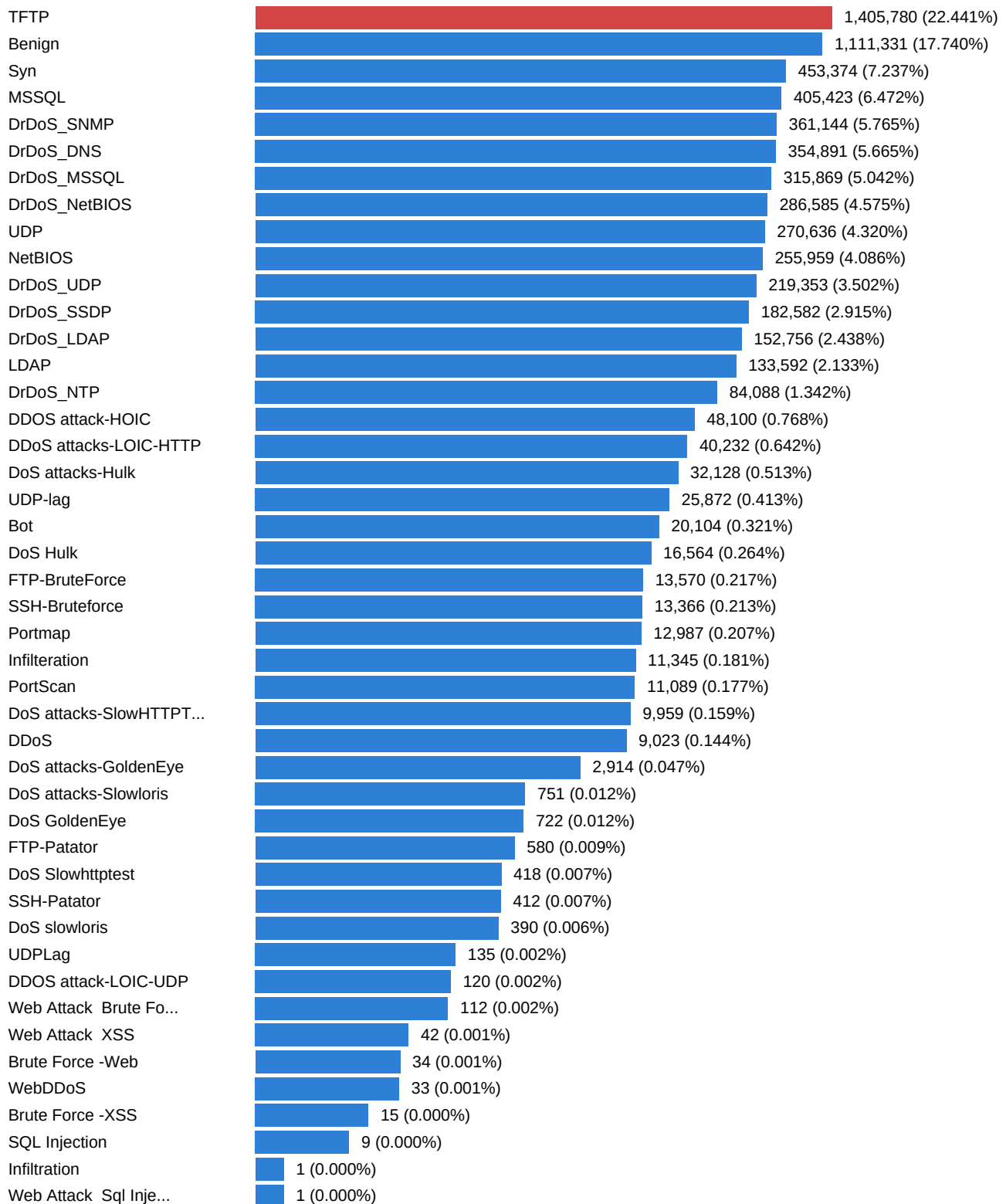
Manifest metadata

rows_written: 6264391

output_columns: 88

created_at: 2026-05-09T10:37:43

Label Distribution (log-scaled)



Where the Imbalance Happens

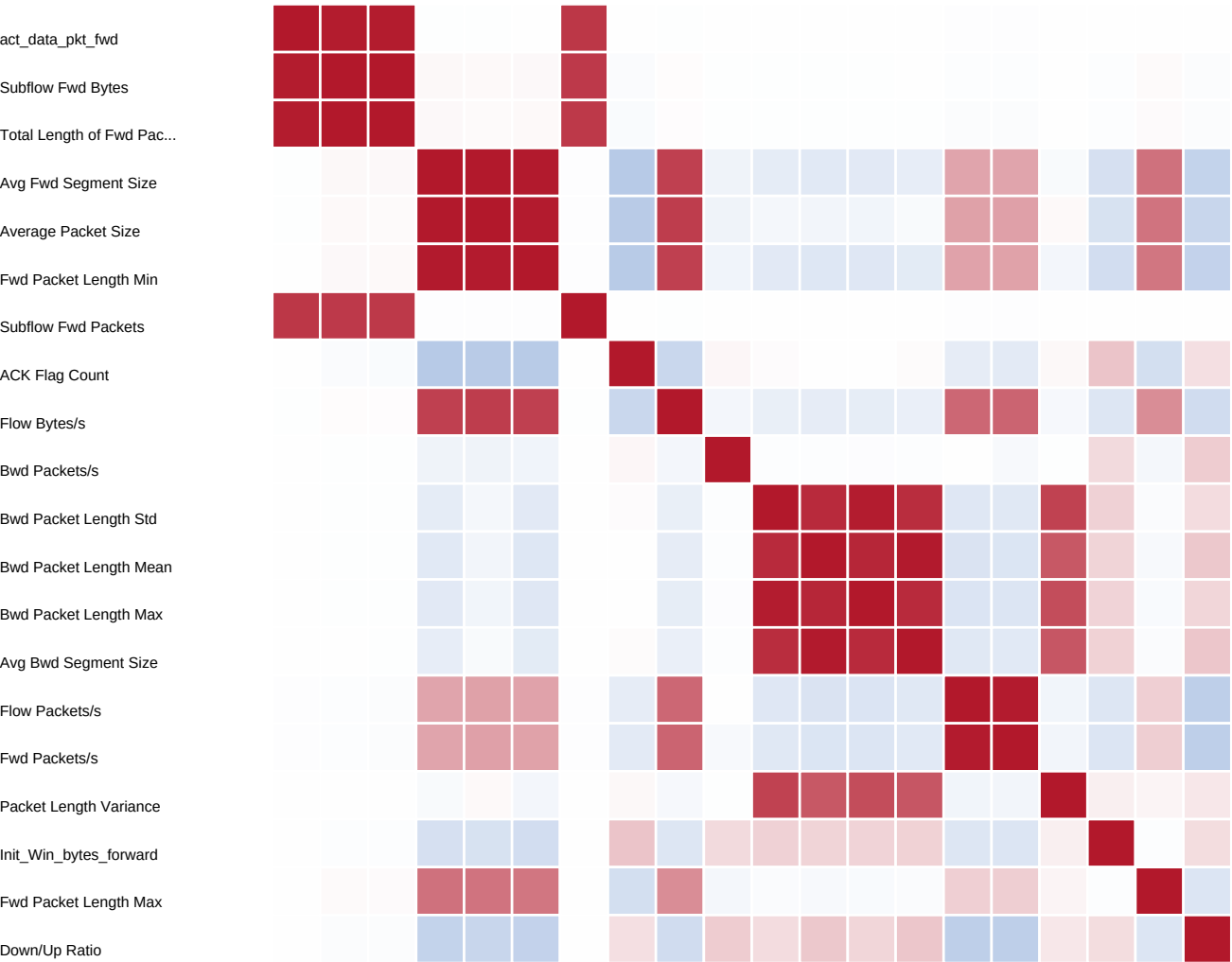
Smallest classes

Label	Count	Percent	Ratio to majority
Infiltration	1	0.00002%	1:1405780.0
Web Attack Sql Injection	1	0.00002%	1:1405780.0
SQL Injection	9	0.00014%	1:156197.8
Brute Force -XSS	15	0.00024%	1:93718.7
WebDDoS	33	0.00053%	1:42599.4
Brute Force -Web	34	0.00054%	1:41346.5
Web Attack XSS	42	0.00067%	1:33471.0
Web Attack Brute Force	112	0.00179%	1:12551.6
DDOS attack-LOIC-UDP	120	0.00192%	1:11714.8
UDPLag	135	0.00216%	1:10413.2
DoS slowloris	390	0.00623%	1:3604.6
SSH-Patator	412	0.00658%	1:3412.1
DoS Slowhttptest	418	0.00667%	1:3363.1
FTP-Patator	580	0.00926%	1:2423.8
DoS GoldenEye	722	0.01153%	1:1947.1
DoS attacks-Slowloris	751	0.01199%	1:1871.9
DoS attacks-GoldenEye	2,914	0.04652%	1:482.4
DDoS	9,023	0.14404%	1:155.8
DoS attacks-SlowHTTPTest	9,959	0.15898%	1:141.2
PortScan	11,089	0.17702%	1:126.8
Infiltration	11,345	0.18110%	1:123.9
Portmap	12,987	0.20731%	1:108.2

Interpretation

The imbalance is concentrated in classes whose percentage is below 1 percent of the dataset and especially below 0.1 percent. During training, these classes can be under-learned because weighted aggregate scores are dominated by high-support labels.

Feature Correlation Heatmap



Selected feature order

1. act_data_pkt_fwd
2. Subflow Fwd Bytes
3. Total Length of Fwd Packets
4. Avg Fwd Segment Size
5. Average Packet Size
6. Fwd Packet Length Min
7. Subflow Fwd Packets
8. ACK Flag Count
9. Flow Bytes/s
10. Bwd Packets/s
11. Bwd Packet Length Std
12. Bwd Packet Length Mean
13. Bwd Packet Length Max
14. Avg Bwd Segment Size
15. Flow Packets/s
16. Fwd Packets/s
17. Packet Length Variance
18. Init_Win_bytes_forward
19. Fwd Packet Length Max
20. Down/Up Ratio

Method Notes and Recommendations

- Label counts are exact because the script scans the Label column across the full CSV.
- The heatmap is sample-based to keep memory usage stable on a multi-gigabyte CSV. Increase `--sample-rows` if the machine has more RAM.
- For modelling reports, do not rely only on weighted precision/recall/F1. Weighted metrics can look strong when majority classes dominate. Add per-class recall/F1 for the rare labels listed above.
- Suggested mitigations: stratified train/test split, class weights, focal loss or re-sampling, and threshold tuning for rare attack classes.
- Generated files include SVG plots, CSV tables, JSON summary, HTML report, and this PDF report.